

AI Summer School 2026
Medical Imaging Informatics

University of Pittsburgh

**Foundations of Explainable
AI in Medical Imaging**

Instructor: Chloe Fox, MSHI

Agenda

- Where are we with Artificial Intelligence (AI)?
- **Trustworthy AI:**
 - **AI Responsibility (R)**; Why? What?
 - **AI Explainability (E)**; Why? What? **Uncertainty** Quantification.
 - **AI Fairness (F)**; Why? What?
- Principles for **Trustworthy AI** or/and **REF-AI** in Medical Imaging
 - **REF** (Responsible, Explainable, Fair)

Learning Objectives

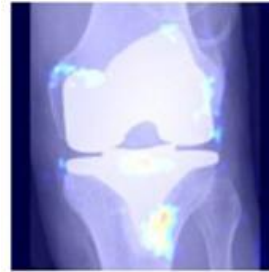
After completing this module, you should be able to:

- Understand and explain where we are with AI in Medical Imaging
- Understand medical imaging informatics
- Discuss AI Explainability in Medical Imaging
- Discuss AI Responsibility in Medical Imaging
- Discuss AI Fairness in Medical Imaging

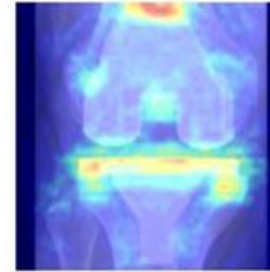


Where are we with AI?

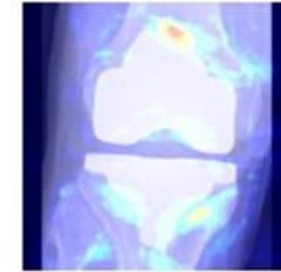
Knee Radiographs; AP View



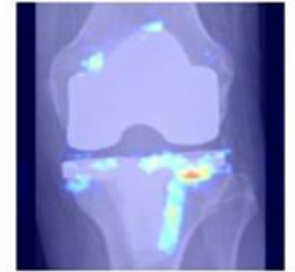
(a) Attune



(b) Sigma



(c) Triathlon

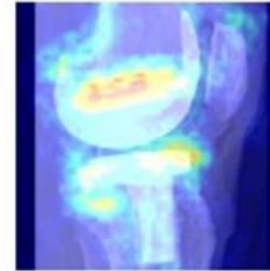


(d) Persona

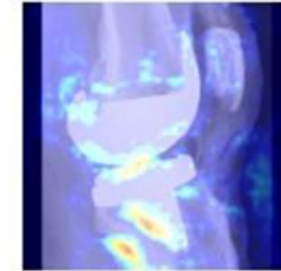
Knee Radiographs; Lateral View



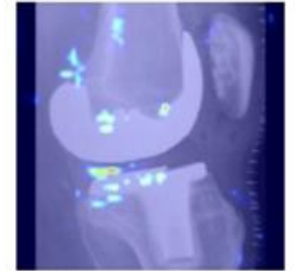
(e) Attune



(f) Sigma



(g) Triathlon



(h) Persona

Clinical Significance:

- TJA is the fastest growing surgery nationwide
- Revision Surgery

Results: The deep learning (AI) algorithm achieved **the same or higher performance** than **two surgeons** in identifying **four of four TKR implant designs**.

[Home](#) > [Advances in Visual Computing](#) > Conference paper

DeepTKAClassifier: Brand Classification of Total Knee Arthroplasty Implants Using Explainable Deep Convolutional Neural Networks

[Shi Yan](#) , [Taghi Ramazanian](#) , [Elham Sagheb](#), [Sunyang Fu](#), [Sunghwan Sohn](#), [David G. Lewallen](#),
[Hongfang Liu](#), [Walter K. Kremers](#), [Vipin Chaudhary](#), [Michael Taunton](#), [Hilal Maradit Kremers](#)  & [Ahmad P. Tafti](#) 

Where are we with AI?

Hip Radiographs; AP View

Clinical Significance: Revision Surgery

MEDICAL PHYSICS

The International Journal of Medical Physics Research and Practice

Research Article

Comparing the performance of a deep convolutional neural network with orthopedic surgeons on the identification of total hip prosthesis design from plain radiographs

Alireza Borjali, Antonia F. Chen, Hany S. Bedair, Christopher M. Melnic, Orhun K. Muratoglu, Mohammad A. Mord, Kartik M. Varadarajan ✉



Results: The CNN (AI) achieved the same or higher performance than at least one of the surgeons in identifying eight of nine THR implant designs.

Where are we with AI?

KneeXNet-2.5D: A Clinically-Oriented and Explainable Deep Learning Framework for MRI-Based Knee Cartilage and Meniscus Segmentation

Maimouna Sanogo^{1†}, Fengyi Gao^{2,3†}, Nickolas Littlefield^{2,3†},
Ismaeel A. Siddiqui¹, Luke A. Carlson⁴, Amin Rezaei²,
Pavan Bodanki⁵, Zoe Menezes¹, Abha Kanda¹,
Mehrnaz Abedian¹, George M. Mastorakos⁶, Michael R. Kann³,
Soheyla Amirian⁸, Maedeh Agharazidermani⁹,
Arezo Sarkheyli-Hägele¹⁰, Kasey Harshman¹, Nicole Myers¹,
Hilal Maradit Kremers¹¹, James J. Irrgang¹, Brian J. McGrory¹²,
Adolph J. Yates⁷, Johannes F. Plate^{7*}, Ahmad P. Tafti^{1,2,3*}

¹School of Health and Rehabilitation Sciences, University of Pittsburgh, Pittsburgh, USA.

²Intelligent Systems Program, University of Pittsburgh, Pittsburgh, USA.

³Computational Pathology and AI Center of Excellence (CPACE), University of Pittsburgh, Pittsburgh, USA.

⁴School of Medicine, University of Pittsburgh, Pittsburgh, USA.

⁵School of Computing, University of Georgia, Athens, USA.

⁶Scienza Health, Scottsdale, USA.

⁷Department of Orthopaedic Surgery, University of Pittsburgh, Pittsburgh, USA.

⁸Seidenberg School of Computer Science and Information Systems, Pace University, New York, USA.

⁹School of Information, Florida State University, Tallahassee, USA.

¹⁰Department of Computer Science and Media Technology, Sustainable Digitalization Research Center (SDRC), Malmö University, Malmö, Sweden.

¹¹Department of Orthopedic Surgery, Mayo Clinic, Rochester, USA.

¹²Department of Orthopaedic Surgery, Tufts University, Medford, USA.



KneeXNet-2.5D: an AI Tool set for Knee Cartilage and Meniscus Segmentation in MRIs

Upload either:

- A `.npy` file containing MRI slices, OR
- A folder (multiple files) of `.dcm` DICOM images

Upload NPY or DICOM files



Drag and drop files here

Limit 200MB per file • NPY, DCM

Browse files

Please upload a file or a folder of DICOM images.

Where are we with AI?



Please extract drug names from this text:

Non steroidal anti-inflammatory drugs (NSAIDs) and prophylactic radiotherapy can prevent ectopic bone formation around the hip after total hip arthroplasty. We retrieved from Medline Embase and the Cochrane Register (clinical) trials and other relevant literature on the prevention of heterotopic ossification (HO) from 1990-2002 for further review. Review of these clinical trials shows that HO is effectively prevented by a postoperative NSAID treatment with indomethacin for at least seven days. The best evidence is available for indomethacin although naproxen, diclofenac and ibuprofen are also well-documented. Short term use of ibuprofen is not effective.



January 3rd, 2025



The drug names mentioned in the text are:

- Non steroidal anti-inflammatory drugs (NSAIDs)
- Indomethacin
- Naproxen
- Diclofenac
- Ibuprofen

Where are we with AI?

Question:

Ahmad arrived at the cinema before Donna but after Ben. Patti came after Donna but before Ahmad. Who arrived first?

Ben < Ahmad < Donna < Patti
But this violates "Patti < Ahmad".

Ben < Ahmad < Patti < Donna
But this violates "Patti > Donna"

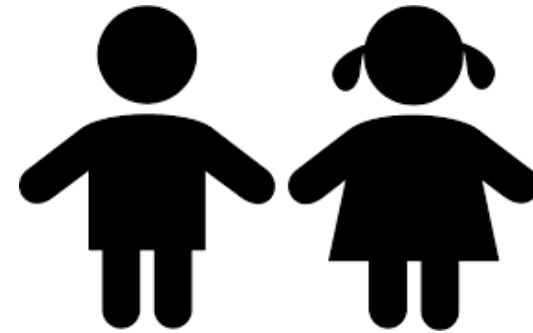


The question itself is valid in structure and sounds natural, but the temporal constraints inside it are logically **inconsistent**, which makes it impossible to answer.

AI; Extremely **Smart** or Strangely **Ignorant**!



Ortho Surgeons



7 Year Old Kids

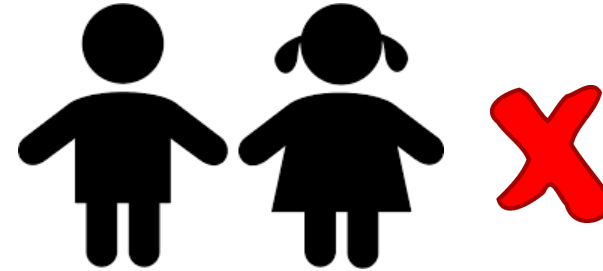


Remember: AI mimics intelligence and does not truly understand the world!

Trustworthy AI: AI Responsibility; Why?

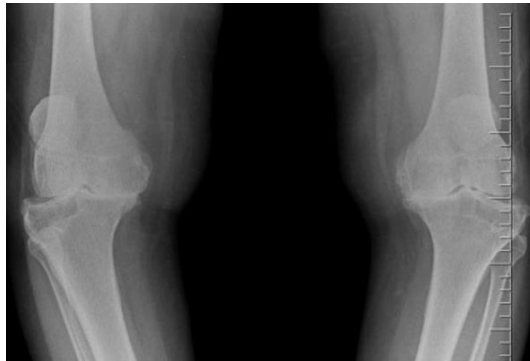
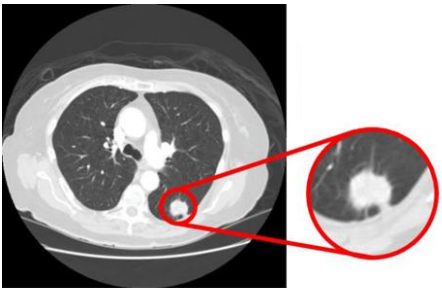


Ortho Surgeons



7 Years Old Kids

What if AI in medical imaging grabs a cancerous lung nodule as non-cancerous?!!! What will be the consequences?

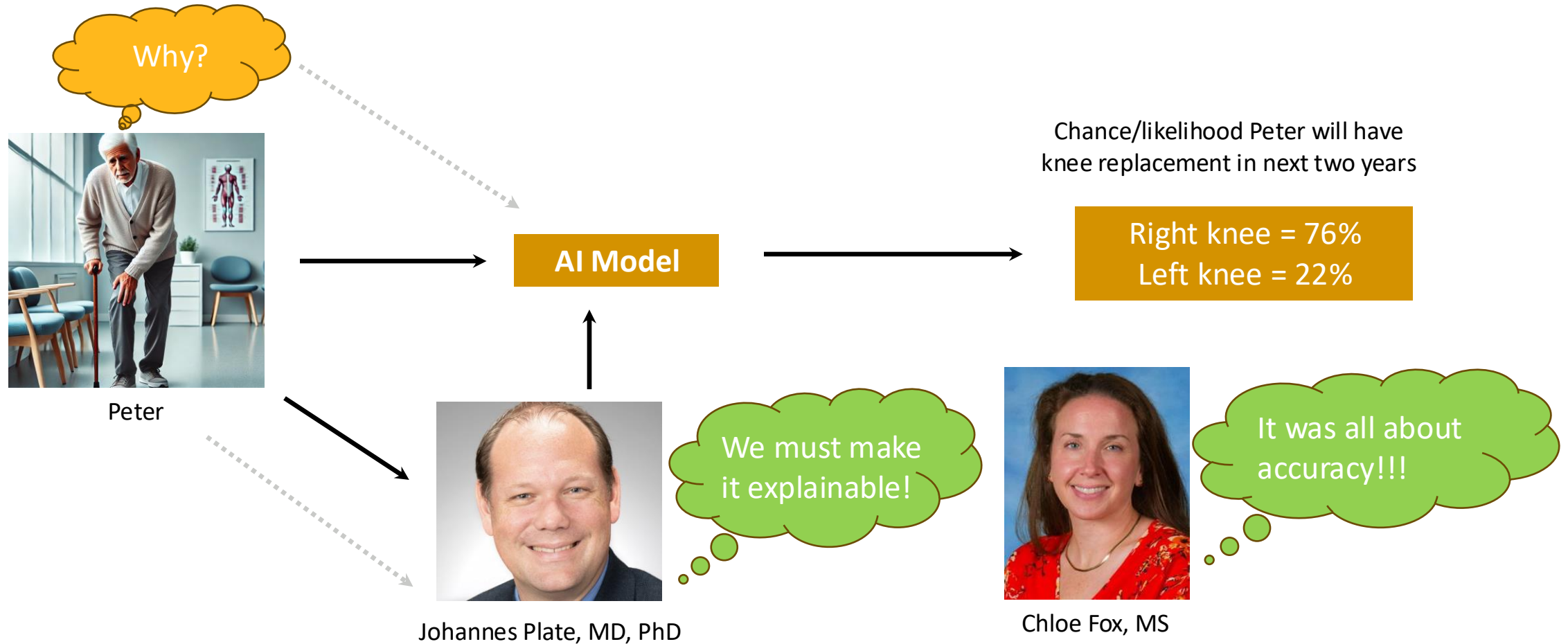


What if AI in medical imaging says this knee does not need replacement (TKA)?!!!

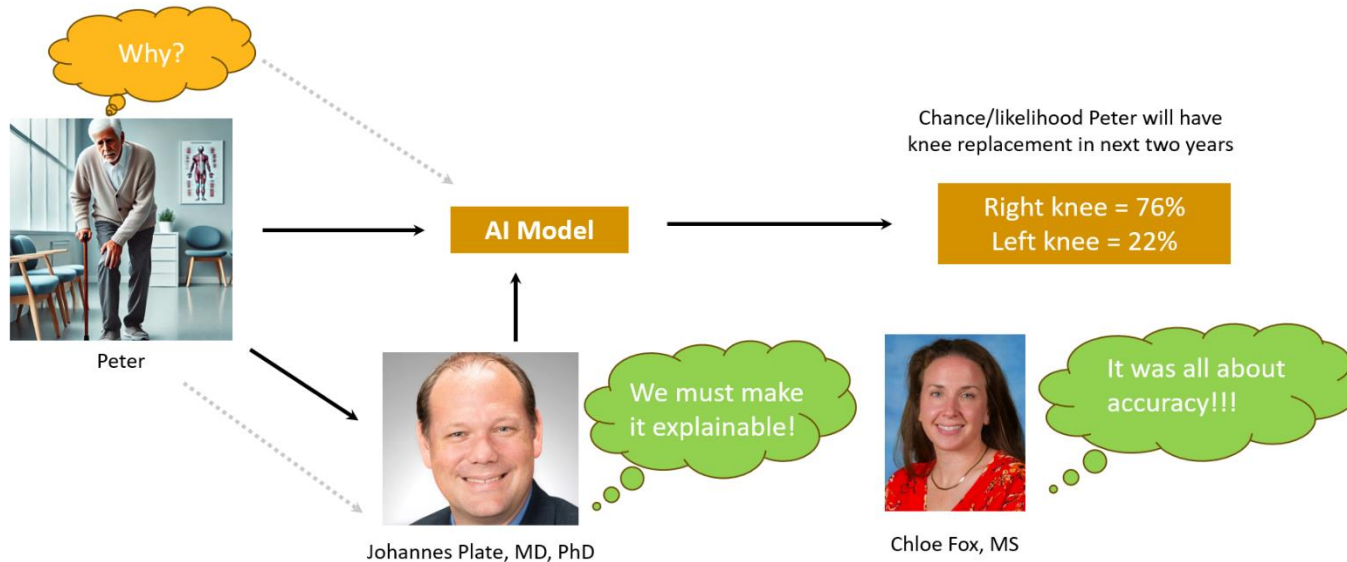
Trustworthy AI: AI Responsibility; What?

- **Responsible AI** refers to building, implementing, and utilizing AI systems in a way that is **ethical, sustainable, and trustworthy**.
- **Responsible AI** in medical imaging ensures that AI tools are implemented and used in ways that align with **medical ethics, trustworthy, fairness and equity, regulations, patient privacy, and safety standards**.

Trustworthy AI: AI Explainability; Why?



Trustworthy AI: Uncertainty-Aware AI; Why?



Scenario:

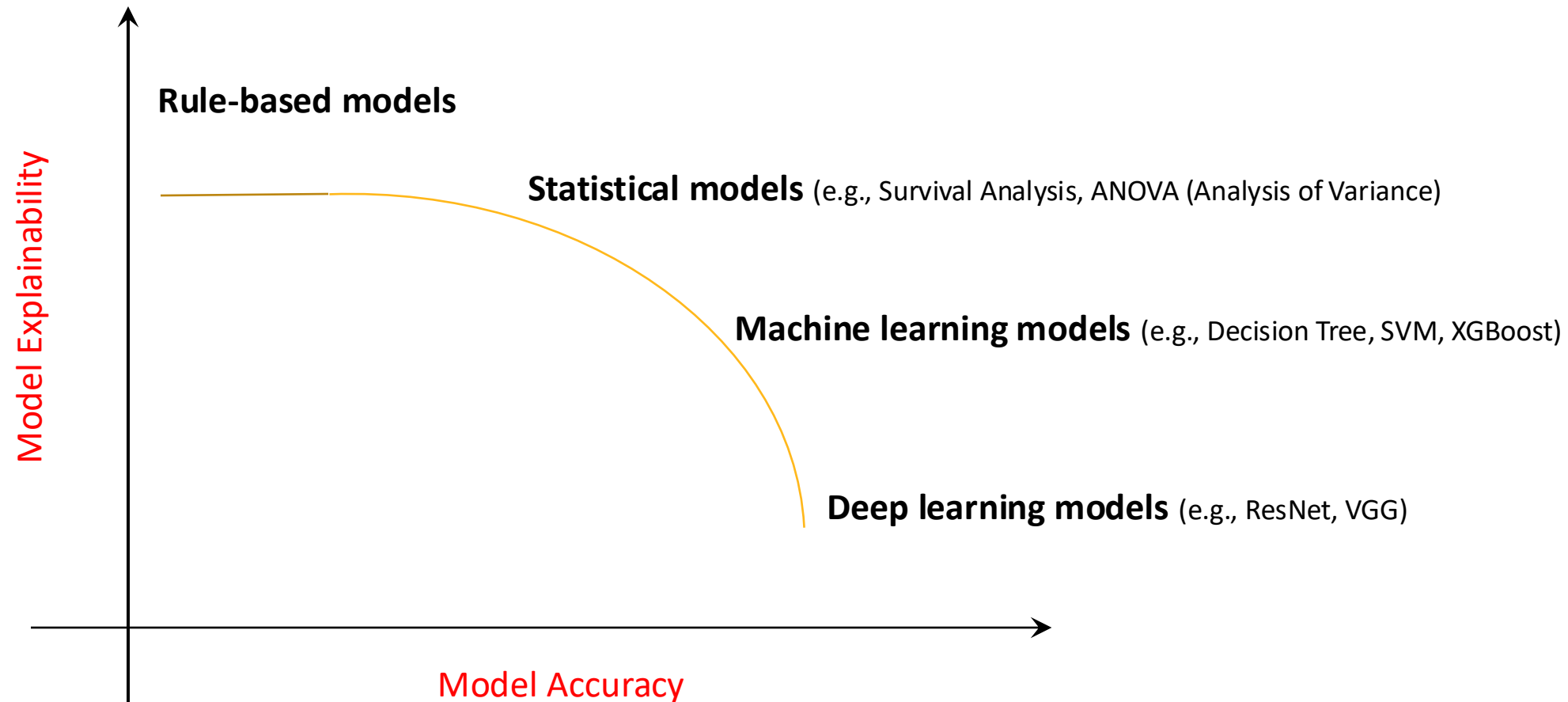
- Chloe will simplify the AI model, so the model will only say **[No surgery needed; Surgery needed]**
- The AI model predicts “**No surgery needed**” for Peter’s right knee for the very next year.
- However, this was the AI model output **[0.52; 0.48]**
- Class 0 = **"No surgery needed"**
- Class 1 = **"Surgery needed"**

Problem:

The surgeon (Dr. Plate), unaware of the low certainty, accepts the decision at face value. Months later, Peter returns in pain, his condition worse, and now requiring more complex and riskier surgery.

Trustworthy AI: AI Explainability; Why?

Explainability VS Accuracy



Trustworthy AI: AI Explainability; What?

- **AI explainability** in healthcare refers to the ability of AI models to provide clear and **understandable explanations of their decisions** and **predictions** to their end-users, such as **clinicians, surgeons, and patients**.
- In medical image analysis, **explainable AI** is critical because healthcare professionals must trust and understand the **AI-powered imaging toolsets** before any utilization for patient care and clinical practices.

Trustworthy AI: Uncertainty-Aware AI; What?

- **Uncertainty-aware AI** refers to the **ability of AI models** to not only generate predictions but also quantify and communicate **how confident they are in those predictions**.
- **In medical imaging**, uncertainty estimation is critical because clinicians must understand whether an AI's output is **reliable or uncertain** before applying it to a patient's care plan
 - Uncertainty-aware AI doesn't just say *what it sees* — it also says *how **sure** it is about what it sees*.

AI Fairness; Why?

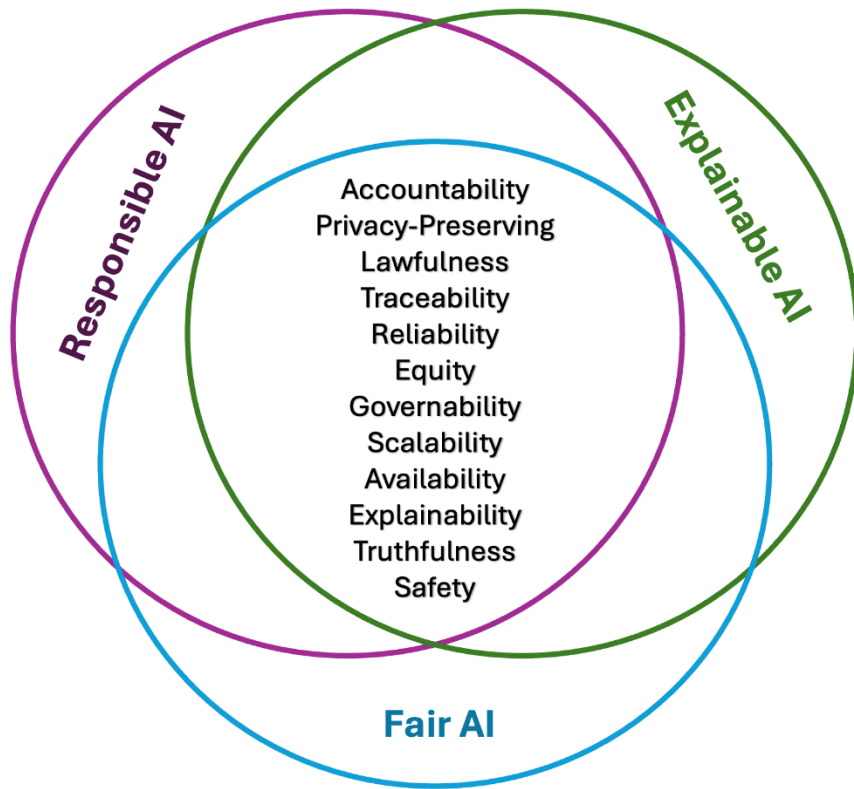


Kurt R. Weiss, MD
Associate Professor
Director, Musculoskeletal Oncology Laboratory
Vice Chair of Translational Research
Department of Orthopaedic Surgery, University of Pittsburgh
UPMC Hillman Cancer Center

Trustworthy AI: AI Fairness; What?

- **AI fairness** in medical imaging ensures that **AI algorithms** are designed and trained to provide **equitable outcomes** for all **demographic groups**, including **race, gender, age, socioeconomic status**, and **geographic location**.
 - This prevents/mitigates **biased** or unequal performance in detecting medical conditions across different populations.

Principles for REF-AI in Medical Imaging (Contd.)

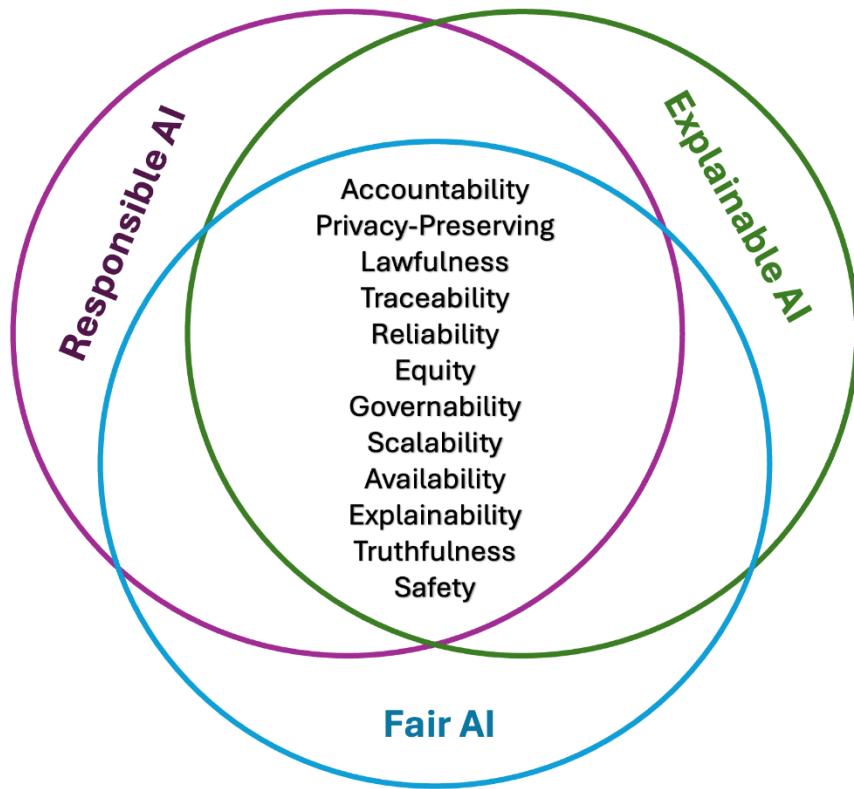


Accountability: It ensures that AI systems are transparent, auditable, and compliant with defined legal and ethical standards, holding developers and organizations responsible for system performance and outcomes.

Privacy-preserving: It refers to developing and deploying AI systems that safeguard sensitive patient information while maintaining the system's ability to deliver reliable, trustworthy, accurate, and actionable insights.

Lawfulness: It refers to aligning AI systems with established legal and regulatory frameworks, guaranteeing adherence to norms and standards that protect patients and uphold ethical principles.

Principles for REF-AI in Medical Imaging (Contd.)

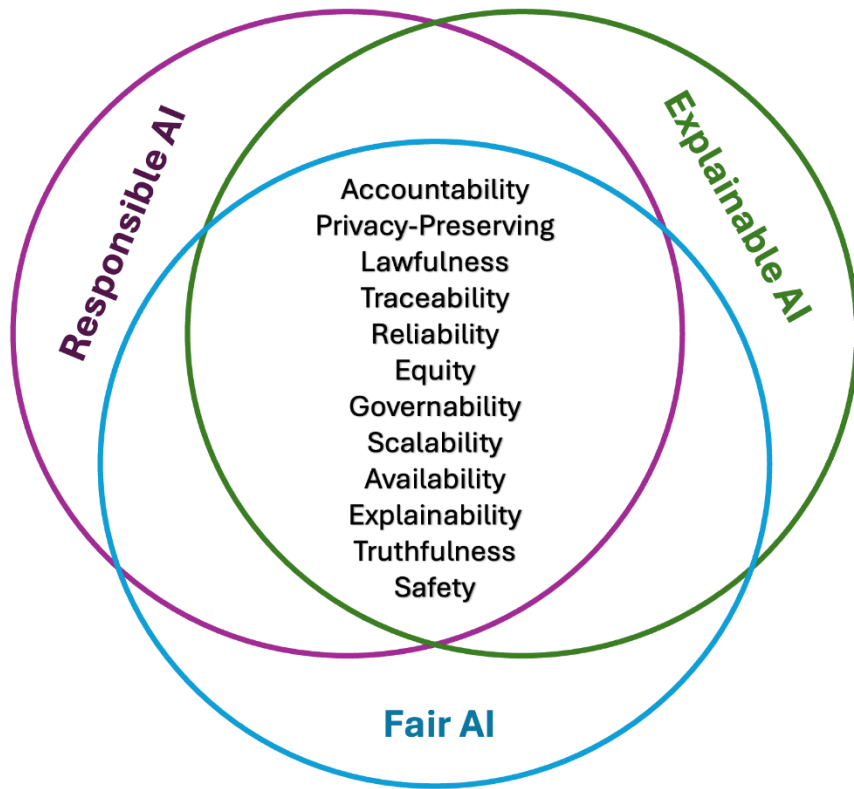


Traceability: It involves maintaining a comprehensive record of each phase in the system's life cycle, starting from functional and nonfunctional requirement analysis, system specification, and data collection to AI model development, validation, and deployment

Reliability: It means consistently performing as expected across diverse healthcare settings and scenarios, managing uncertainties and errors, and preventing breakdowns or performance issues to maintain core functionalities.

Equity: It addresses fair access, representation, and outcomes across diverse populations.

Principles for REF-AI in Medical Imaging (Contd.)

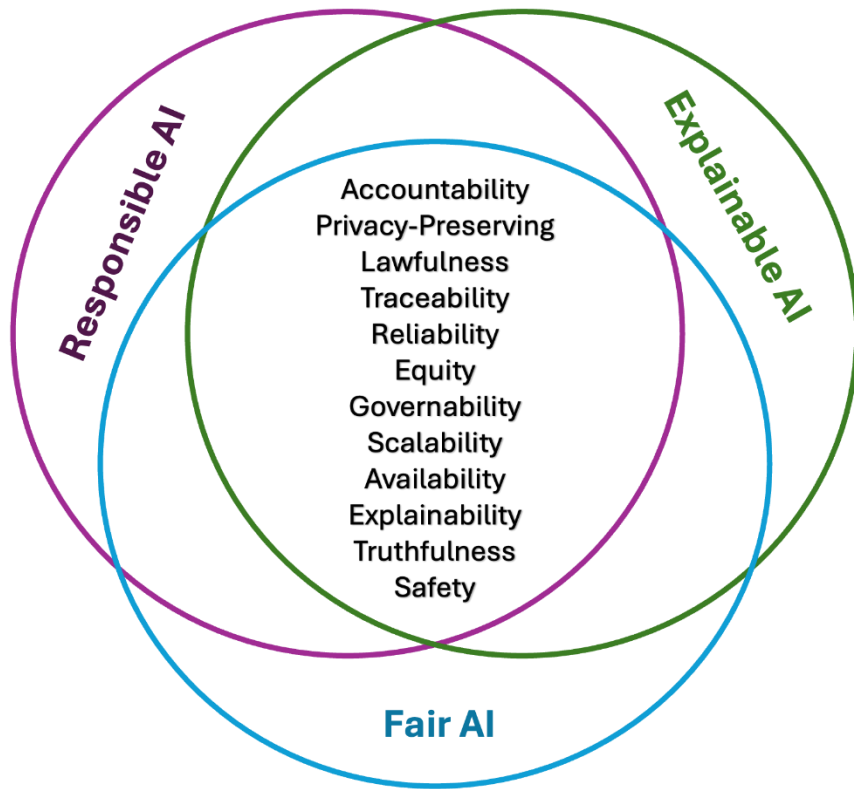


Governability: It is critical to maintaining accountability, upholding ethical standards, and building trust within healthcare systems. The rapid integration of AI in healthcare and medical imaging analysis begs a robust governance model that addresses ethical and practical challenges.

Scalability: It refers to the capacity of AI systems to efficiently handle growing workloads and user numbers and maintain consistent performance as they expand.

Availability: aims to build AI tool sets that are easy to use and accessible for all types of healthcare providers, regardless of technology or data system's level, making them available for all parties.

Principles for REF-AI in Medical Imaging



Explainability: clarifies how AI systems make decisions, process data and images, and function internally.

Truthfulness: It means the ability of AI systems to provide accurate, consistent, and fact-based outputs aligned with medical knowledge and clinical evidence..

Safety: it involves building and implementing AI systems that are robust, dependable, and free from behaviors or predictions that could lead to harm.

State-of-the-Art in Responsible, Explainable, and Fair AI for Medical Image Analysis



Received 29 January 2025, accepted 19 March 2025, date of publication 27 March 2025, date of current version 8 April 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3555543



State-of-the-Art in Responsible, Explainable, and Fair AI for Medical Image Analysis

**SOHEYLA AMIRIAN¹, FENGYI GAO²,
NICKOLAS LITTLEFIELD^{3,4}, (Graduate Student Member, IEEE),
JONATHAN H. HILL¹, ADOLPH J. YATES JR.⁵, JOHANNES F. PLATE⁵,
LIRON PANTANOWITZ^{4,6}, HOOMAN H. RASHIDI^{4,6},
AND AHMAD P. TAFTI^{1,2,3,4,6}**

¹Seidenberg School of Computer Science and Information Systems, Pace University, New York, NY 10038, USA

²Department of Health Information Management, University of Pittsburgh, Pittsburgh, PA 15260, USA

³Intelligent Systems Program, University of Pittsburgh, Pittsburgh, PA 15260, USA

⁴Computational Pathology and AI Center of Excellence (CPACE), University of Pittsburgh School of Medicine, Pittsburgh, PA 15261, USA

⁵Department of Orthopaedic Surgery, University of Pittsburgh, Pittsburgh, PA 15213, USA

⁶Department of Pathology, University of Pittsburgh School of Medicine, Pittsburgh, PA 15261, USA

Corresponding authors: Soheyla Amirian (samirian@pace.edu) and Ahmad P. Tafti (tafti.ahmad@pitt.edu)



Questions to ponder

Responsible: Clinical Accountability

- If an AI model misclassifies a malignant lesion and a patient suffers harm, who is responsible — the radiologist, the hospital, the developer, or the regulator?
 - Does responsibility change if the model was FDA-approved?

Explainable: Cognitive Burden and Patient Right to Know

- Could too much explainability overwhelm clinicians?
- Do patients have a right to know that AI was involved in their care?

Fairness: Bias and Performance

- Can AI trained on historically biased healthcare data ever truly be fair?
 - Are we encoding structural inequities into models?
 - Should we sacrifice performance for fairness?